



Xi'an Jiaotong-Liverpool University

西交利物浦大學

IOM209 BUSINESS INTELLIGENCE

INDIVIDUAL REPORT

2023/2024 SEMESTER 2

BSC INFORMATION MANAGEMENT AND INFORMATION SYSTEMS

YUNZHOU ZHONG

2145702

MAY 20, 2024

Contents

1	Introduction	2
2	Literature Review	3
2.1	Behavioral Inhibition and Approach System	3
2.2	Financial Market Conditions Affect Information Attention	3
2.3	Research Gap and Objectives	4
3	Hypothesis Development	4
4	Methodology	5
4.1	Data	5
4.2	Variables and Measurement	6
4.2.1	Dependent Variable	6
4.2.2	Independent Variables	6
4.3	Study 1	6
4.4	Study 2	6
4.5	Study 3	7
4.5.1	Machine Learning Algorithms	7
4.5.2	Feature Analysis	8
5	Discussion	10
6	Conclusion	11
6.1	Limitations and Future Research	11
6.2	Brief Conclusion	11
	Refences	12
	Appendices	14

Still Attention? Investment Returns and Retail Investor Market Attention Behavior, Evidence from CSI 300

Abstract

In China's stock market, there is a very special problem, retail investors are the largest group of investors, this study examines the impact of investment returns on retail investors' market attention behavior, particularly the role of the Behavioral Inhibition System (BIS) and Behavioral Approach System (BAS) under different market conditions. The study analyzes the impact of different investment returns on market attention using CSI 300 index constituent data from 2014 to 2016, combining statistical methods and advanced machine learning techniques. The study's results validate three hypotheses: first, investment returns significantly affect investors' behavioral inhibition, and negative returns cause investors to pay less attention to stock information. Second, when investment returns are consistently negative or the market as a whole is depressed, investor attention fluctuates as a result of the alternating roles of the BIS and the BAS systems. Lastly, the machine learning model effectively predicts and explains the impact of return changes on investor behaviors, especially the relative impact of BIS and BAS systems on different behavioral indicators. These results not only expand the theoretical framework of behavioral finance and behavior-related information systems research but also provide financial institutions and market regulators with new ways to understand and predict the behavior of market participants to optimize investment strategies and market regulatory measures.

1 Introduction

The democratization of finance, fueled by the internet, has significantly increased market participation, especially among previously underrepresented socioeconomic groups (Hvide et al., 2024). This transformation has facilitated a more diverse array of market participants, who bring with them varying degrees of sensitivity to emotional returns. Emotional returns, reflecting the positive or negative emotional impacts of investment gains or losses, are critical in influencing investor behavior and market attention, making their study essential in the field of behavioral finance.

Two key psychological systems influence investors' reactions to market stimuli: the Behavioral Inhibition System (BIS) and the Behavioral Approach System (BAS). The BIS is activated by negative stimuli, leading to avoidance behaviors, whereas the BAS promotes approach behaviors in response to positive stimuli. This framework is instrumental in analyzing how different types of emotional returns affect investor behaviors, particularly in volatile markets (Muehlfeld, Weitzel and Witteloostuijn, 2013).

Significant research highlights that media sentiment, especially pessimism, can lead to market declines, which eventually revert to align with fundamental values, underscoring the impact of emotional sentiment on market dynamics (Tetlock, 2007). Moreover, the observable behaviors of investors, such as internet searches and social media engagement, act as precursors to market movements, offering crucial insights for market participants and policymakers (Chai et al., 2021). Search volume can also be used as a proxy for retail investor attention metrics to study the relationship between returns and attention (Wang et al., 2018).

This study utilizes a blend of statistical methods to establish significant correlations and advanced machine learning techniques to predict behaviors. It specifically examines how different types of returns influence various aspects of market attention,

including participation in stock-related forums, specific stock code searches, and comprehensive stock information searches(Li et al., 2024).

Focusing on the Chinese financial market during the volatile period of 2014 to 2016, this study leverages data from the CSI 300 index components. This period, encompassing significant market events like the 2015 stock market crash, offers a robust dataset for examining the impact of emotional shifts on market attention. The study aims to understand how the unique characteristics of the Chinese market, which features a high proportion of sentiment-driven retail investors, affect the relationship between emotional returns and market behavior (Reyes, 2019). Because retail investors are very active in acquiring information and responding to market changes (Jin and Xue, 2023). This has been one of the focal topics of China's stock market behavior in recent years.

Using Chinese market data from 2014 to 2016, this study focuses on the impact of major market events, such as the 2015 stock market crash, on changes in sentiment during periods of market turbulence, and explores how these changes affect market attention(Zhao, Chen and Zhang, 2019). The Chinese market is characterized by a high proportion of sentiment-driven retail investors, which provides a unique perspective to study the relationship between sentiment returns and market behavior. Through a combination of econometric analysis and machine learning, this study makes progress in understanding how emotional returns shape investor behavior and market attention, especially in an emerging market environment.

2 Literature Review

2.1 Behavioral Inhibition and Approach System

Gray (1982) proposed the Behavioral Inhibition System (BIS) and Behavioral Activation System (BAS) theories. They are now widely used to understand how an individual's sensitivity to punishment and reward affects his or her behavior. In recent years, the theoretical frameworks of these two systems have been used to analyze investor behavior in financial markets, especially in environments of uncertainty and potential risk. Personality traits such as BIS and BAS are predictive of investor cognitive biases, where individuals high in BIS are likely to be more cautious, while individuals high in BAS may be more willing to take risks (Muehlfeld, Weitzel and Witteloostuijn, 2013). There is a complex interaction between retail investors' attention and market behavior, where the activity levels of BIS and BAS may influence their decision-making process (Jin and Xue, 2023). Changes in stock market volatility and trading volume also reflect how investors' BIS and BAS are mobilized. BIS and BAS influence users' behavioral intentions and usage dynamics. In particular, people with higher BIS tend to pay less attention when performance is erratic or does not meet expectations (Aydinliyurt et al., 2021). The psychological profile of investors is closely related to their decision-making behavior, in which the level of BIS and BAS directly affects their response to risk and return. Individuals with high BIS tend to avoid risky investments, while individuals with high BAS search for high return opportunities (Muehlfeld, Weitzel and Witteloostuijn, 2013). Most studies now examine how BIS and BAS affect investor behavior, and there is little on the performance of investor activities under the influence of BIS and BAS.

2.2 Financial Market Conditions Affect Information Attention

Psychological and Neuroscience Factors Affect Financial Decision Making (Frydman and Camerer, 2016). Investors are influenced by a variety of psychological and neuroscientific factors when making financial decisions, with factors such as emotion and attention playing an important role. Interactions on social media significantly affect investors' attention and trading behavior. In the Internet trading environment, investor attention is more likely to be attracted to market price trends (Xian Li and Teall, 2016). Sentiments expressed on social platforms such as microblogs directly affect stock market performance, where positive sentiments are associated with stock market gains (Sun et al., 2020). There is a significant interaction between Microblog Sentiment and Stock Returns, especially in the presence of negative sentiment (Deng et al., 2018).

While the Internet has attracted widespread investor attention, this attention is not always based on company fundamentals (Zhang and Zhang, 2015). Investors' decision-making behavior is significantly influenced by digital notifications such as

push messages (Arnold, Pelster and Subrahmanyam, 2022). During market turbulence, investor attention is limited, which affects the processing of market information and stock returns (Zhu et al., 2020). The use of investor attention metrics, such as the volume of Internet searches, can significantly improve the accuracy of predicting stock price movements (Zhang, Chu and Shen, 2021). The immediate dissemination of information affects stock prices through channels such as social media (Agarwal, Kumar and Goel, 2019).

2.3 Research Gap and Objectives

Although there has been considerable research on investor behavior, there is still a dearth of research on the specific behavioral patterns of retail investors under different market conditions, especially in the context of the impact of returns on sentiment. While machine learning has been widely used in financial forecasting, its application in explaining and predicting investor behavior has not gone far enough, especially when combined with psychological theories.

The following research and hypotheses will provide insights into the behavioral patterns of retail investors under different market return conditions, especially the impact of affective returns on their market attention behavior. It also predicts the impact of changes in investment returns on investor behavior in conjunction with machine learning methods, and will explain the importance of different features in the prediction.

3 Hypothesis Development

BIS and BAS play an important role in explaining investor behavior. Investors with high BIS levels tend to reduce risk exposure and information processing during market instability (Muehlfeld, Weitzel and Witteloostuijn, 2013). This behavior can be viewed as a psychological defense mechanism aimed at reducing potential losses. However, as market volatility and uncertainty increase, investors' return expectations may suffer, further reinforcing their behavioral disinhibition tendencies. Specifically, when investors perceive lower investment returns or poor market performance, they may pay less attention to stock information (Drake et al., 2017). This reduced attention behavior is not only to avoid the psychological stress of information overload but also to reduce the risk of making wrong decisions. In addition, the influence of market sentiment and digital media information can exacerbate this phenomenon by making investors more inclined to rely on emotions and intuition rather than rational analysis (Richards et al., 2018). Therefore, the hypothesis is proposed:

H1: Influenced by investment returns, investors may develop behavioral inhibitions and pay less attention to stock information.

In a market environment, sustained negative investment returns have a significant impact on investor behavior. Based on the Behavioral Inhibition System (BIS) and Behavioral Activation System (BAS) theories, when investment returns are persistently negative, the BIS causes investors to become more cautious and conservative, and to pay less attention to market information to avoid further losses. However, this dampening effect is not sustained. When market performance is chronically low, BAS may motivate investors to seek new opportunities or engage in risk management in the hope of reversing negative returns (Daniel, Hirshleifer and Teoh, 2002). Thus, the alternating effects of BIS and BAS can lead to fluctuations in investor attention. Therefore, hereafter, the hypothesis is proposed:

H2: When investment returns continue to be negative or the market as a whole is depressed, investors' attention fluctuates due to the alternating roles of BIS and BAS systems.

Based on the first two hypotheses, recognize that investment returns have a significant impact on investor behavior and that this impact fluctuates with persistent negative returns (Grinblatt and Keloharju, 2000). To gain a deeper understanding of these behavioral changes, machine learning models will be introduced to predict and explain investor behavioral patterns by analyzing the relationship between changes in returns and different metrics. Therefore, the hypothesis is proposed:

H3: Machine learning models are effective in predicting and explaining the impact of return changes on investor behavior, particularly the relative impact of the BIS and BAS systems on different behavioral metrics.

4 Methodology

4.1 Data

This study concentrates on gathering information from two different sources to analyze Chinese companies that are publicly traded on the Shanghai Stock Exchange and Shenzhen Stock Exchange. Data from 2014-2016 were selected, with the classification and number of listed company Guba posts¹ and listed company web search index² from the China Research Data Service Platform (CNRDS)³, and basic information on listed companies and individual stock returns from the China Stock Market and Accounting Research (CSMAR)⁴ database. The stock code and time are used to match all the indicators. Since the daily frequency data may have a large noise effect, the data are processed by converting the daily frequency data into weekly frequency data and removing abnormal companies such as Special Treatment(ST) and delisted companies, and the data have 382,229 observations.

Due to a large amount of data and the comprehensive consideration of the characteristics of the Chinese stock market, the data is further processed, and we choose to use the constituent stocks of the CSI 300⁵ index for the study because the CSI 300 has a high market capitalization coverage, a high correlation with the market's broader indexes, and a large number of high-quality stocks in the market are concentrated in the sample, which has better representativeness and liquidity (Chu et al., 2023). From 2014 to 2016, a total of 430 stocks have become the CSI 300 constituent stocks, and after screening, the data has 55,136 observations. The detailed data processing flow is shown in Figure 1. In addition, the data overview is presented in Appendix A.

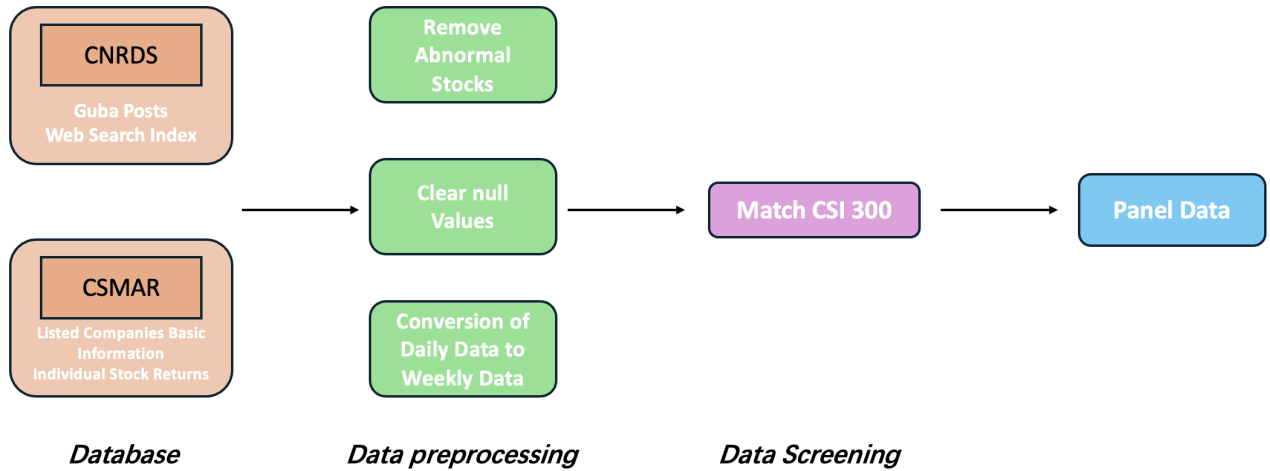


Figure 1: Data Processing Process

¹<https://www.cnrds.com/Home/Index/FeaturedDatabase/DB/GUBA/ViewName/>

²<https://www.cnrds.com/Home/Index/FeaturedDatabase/DB/WSVI/ViewName/>

³<https://www.cnrds.com>

⁴<https://data.csmar.com>

⁵<https://www.csindex.com.cn/#/indices/family/detail?indexCode=000300>

4.2 Variables and Measurement

4.2.1 Dependent Variable

Focus on four potential retail investor attention behaviors, the number of individual stock Guba posts, the number of individual stock Guba comments, the number of individual stock ticker web searches, and the number of individual stock information web searches. These are the four types of statistical information collected, each reflecting market concern behavior in a different sense. The two data from Guba more directly reflect the attention of the more active market participants, but there is noise in this data because there are still a large number of lobbying groups in the Chinese market that put out information to attract retail investors(Chen et al., 2019). The stock code search is more for investors who are more knowledgeable about the market, as the numerical form of ticker symbols is not relatively easy to memorize. The stock information network search index includes the stock code, the company abbreviation, the full name of the company, etc. as the keyword search value sum, this index reflects the more popular attention.

4.2.2 Independent Variables

The dependent variable is set to the weekly rise and fall of the individual stock, with rises labeled as 1 and falls labeled as 0. There are no rises or falls of 0 for the current week in the data, so they are not processed.

4.3 Study 1

This study evaluates the hypothesis of H1: Influenced by investment returns, investors may develop behavioral inhibitions and pay less attention to stock information. To address the role of IV on DV, statistical methods were used to check its significance, through t-test and ANOVA test, it was found that the effect of positive and negative investment returns on the stock information network search index is significant ($T=2.84$, $F=11.23$, $p(t)=0.004526$, $p(F)=0.000805$). The detailed statistical indicators are shown in Table 1. Therefore, it is inferred that H1 is valid and investors may develop behavioral inhibition and pay less attention to stock information due to the influence of investment returns.

Dependent Variable	T	T-test p-value	F	ANOVA p-value
Readnum	-0.299290	0.764720	0.307059	0.579493
Commentnum	0.647291	0.517447	0.430476	0.511758
SVI_Code	-2.829467	0.004664	0.446426	0.504040
SVI_All	2.839083	0.004526	11.230116	0.000805

Table 1: T-test and ANOVA results for various dependent variables

When stock holdings have negative returns, investors may develop behavioral inhibitions and pay less attention to stock information to reduce the anxiety associated with excessive attention. And expect their stock holdings to produce positive returns in the future (Fama and French, 1988).

4.4 Study 2

To test the conjecture of H2: when investment returns are consistently negative or the market is generally depressed, investor attention fluctuates due to the alternating effects of the two systems, BIS and BAS. Three different market situations are used for verification: 70% negative stock returns, 80% negative stock returns, and 90% negative stock returns. By measuring the correlation between IV and DV in different states, it is found that the correlation is weak and negative, which can be proved that there is a change in the attention of retail investors, which may be affected by the alternation of the BIS and BAS, and their attention fluctuates. The correlation indicators are shown in Table 2 and the correlation matrix is shown in Figure 2.

Correlation	90% Negative Returns	80% Negative Returns	70% Negative Returns
Returns and Search Index	-0.028359	0.001181	-0.023442

Table 2: Correlation of returns with the stock search index when the market as a whole is bad

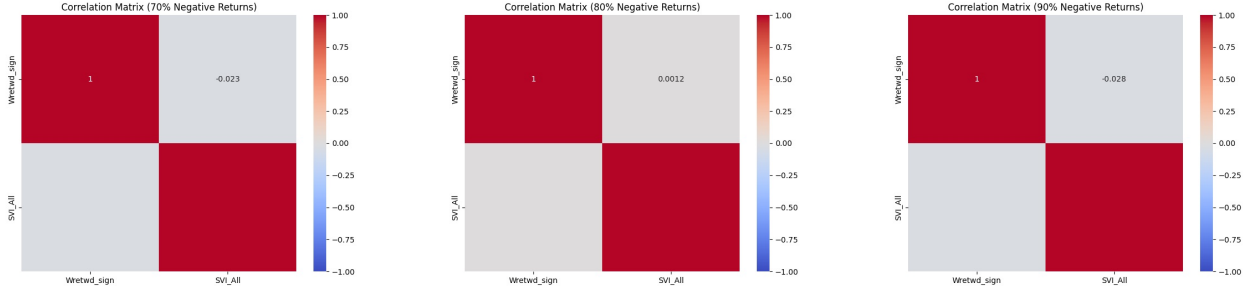


Figure 2: Correlation Matrix

4.5 Study 3

This study uses machine learning to predict changes in different behaviors based on positive and negative yields. Since there is no prediction of stock prices, yields, etc. involved, we do not use time as a control variable and divide 80% of the dataset into training samples and 20% into test samples.

4.5.1 Machine Learning Algorithms

First, the performance of various machine learning models is investigated. Varying the hyperparameters of a model can result in changes to its performance and validity (Yang and Shami, 2020). Consequently, identifying the best hyperparameter settings is crucial (Rijn and Hutter, 2018). This research employed a hybrid approach of grid search and k-fold cross-validation to optimize the hyperparameters of the training model. Initially, a grid search was conducted to outline the optimal range for the model's primary hyperparameters. Following this, k-fold cross-validation (k=5) was employed for hyperparameter selection. The training samples were randomly split into five equal parts, each part serving as a validation set while the remaining four were used as the training set. This process helps in identifying the best hyperparameter settings by training the model under various hyperparameter configurations.

To select the optimal machine learning model, a variety of models were tried to regressively predict the relationship between the four DVs and IV. These models included; XGBoost, AdaBoost, RandomForest, Gaussian Bayes, MLP, and also two neural network models, LSTM and GRU. due to the need to try and account for some of the influences, it was ultimately decided that neural network models would not be used due to their low interpretability (Fodor and Pylyshyn, 1988).

Table 3-5 shows the different evaluation metrics under different DVs, and it can be seen that both MLP and AdaBoost perform well. However, AdaBoost is more stable, therefore, AdaBoost is chosen as the optimal model. For more on model selection, please refer to Appendix B.

Table 3: Table for DV: Readnum

Model	Accuracy	Recall	F1	ROC-AUC
XGBoost	0.518124	0.836135	0.643007	0.507397
AdaBoost	0.524498	1.000000	0.688085	0.510367
RandomForest	0.516829	0.826662	0.641910	0.507009
GaussianNB	0.521182	0.958960	0.677488	0.488462
MLP	0.525042	0.977547	0.683125	0.511738

Table 4: Table for DV: Commentnum

Model	Accuracy	Recall	F1	ROC-AUC
XGBoost	0.520456	0.952359	0.675655	0.506311
AdaBoost	0.524498	1.000000	0.688085	0.511151
RandomForest	0.516777	0.862858	0.651934	0.507332
GaussianNB	0.522140	0.964472	0.679169	0.490580
MLP	0.524446	0.998272	0.687690	0.511145

Table 5: Table for DV: SVI.Code

Model	Accuracy	Recall	F1	ROC-AUC
XGBoost	0.523228	0.860364	0.653997	0.515143
AdaBoost	0.523850	0.963829	0.679620	0.519853
RandomForest	0.520974	0.845740	0.649341	0.513146
GaussianNB	0.521130	0.969974	0.679953	0.510263
MLP	0.523099	0.907721	0.663558	0.521469

Table 6: Table for DV: SVI.All

Model	Accuracy	Recall	F1	ROC-AUC
XGBoost	0.523021	0.920717	0.669402	0.505869
AdaBoost	0.524058	0.993474	0.686467	0.507118
RandomForest	0.522581	0.884038	0.660096	0.507330
GaussianNB	0.524291	0.963507	0.679945	0.513299
MLP	0.524343	0.971326	0.681710	0.513957

4.5.2 Feature Analysis

To visually represent the outcomes of the analysis, a model is developed that combines AdaBoost and SHapley Additive exPlanations (SHAP) to ascertain whether positive or negative returns can be utilized for predicting attention behavior towards retail information. SHAP, which integrates game theory and local explanation, serves as an additive feature attribution technique (Lundberg and Lee, 2017). Furthermore, the utilization of SHAP is widely regarded as a highly effective method for ascertaining fair allocation owing to its characteristics including effectiveness, balance, virtuality, and additivity.(Rodríguez-Pereira et al., 2021). The significance of features is represented by the magnitude of SHAP, while its sign indicates the direction (positive or negative) between independent and dependent variables. A higher magnitude of the SHAP value corresponds to greater significance of the variable in predicting model outcomes (Lundberg and Lee, 2017).

Figure 3 shows the distribution of the impact of each feature in the overall dataset and its corresponding positive and negative changes in stock yields. Most of the high values of the stock code search index (red) correspond to positive SHAP values, verifying that the positive effect of professional investor attention on yields is more pronounced when stock yields are positive. Most of the high values of the number of reads on Guba are also positive, but there is a part of the negative impact, suggesting some complexity in the impact of market attention on yields when yields are positive or negative. The number of comments on Guba's impact is more dispersed, with most of the high values negative, suggesting that a high number of comments may be associated with uncertainty and negative sentiment in the market when yields are negative (Lee, Jiang and Indro, 2002). Stock search index's impact is smaller, with high and low eigenvalues having a limited predictive power for yields, and with an overall trend that is mildly negative, showing more noise in this composite search index when yields are changing.

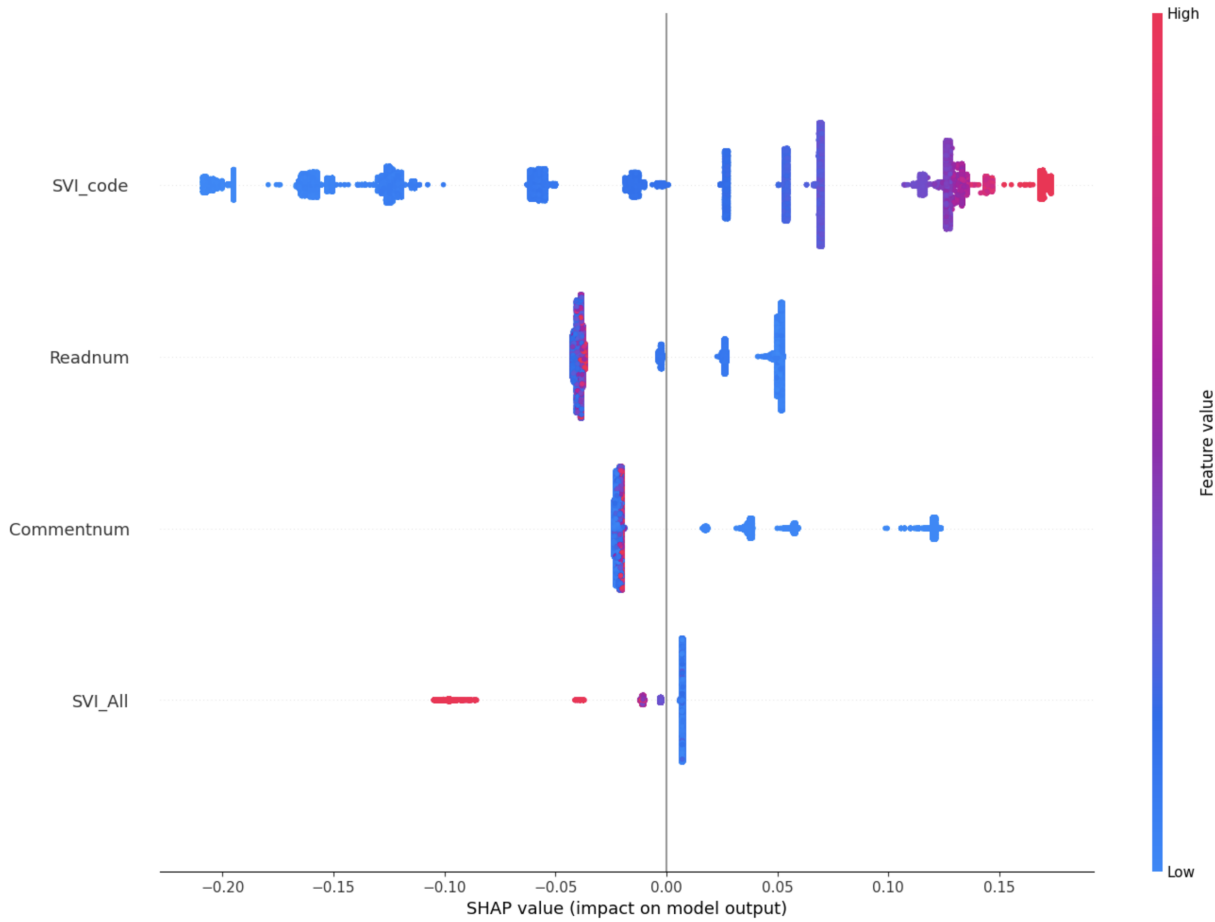


Figure 3: SHAP Values of All Variables

Figure 4 illustrates the effect of each feature on the model output, where changes in feature values correspond to positive and negative changes in stock returns. The number of stock code searches has a significant positive effect on stock returns (+0.13), reflecting the fact that professional investors pay more attention to a particular stock when the return is positive, and vice versa. The number of Guba reads also has a positive effect on returns also has a positive effect (+0.03), reflecting the increased attention of active market participants when returns are positive (Qiu and Welch, 2004). However, number of comments on stock bars has a negative impact on returns (-0.02), probably because a high number of comments implies a

more controversial market, which triggers negative sentiment (Qiu and Welch, 2004). Stock search index has a smaller and slightly negative impact on returns (-0.01), suggesting that this composite indicator contains more noise and uncertainty.

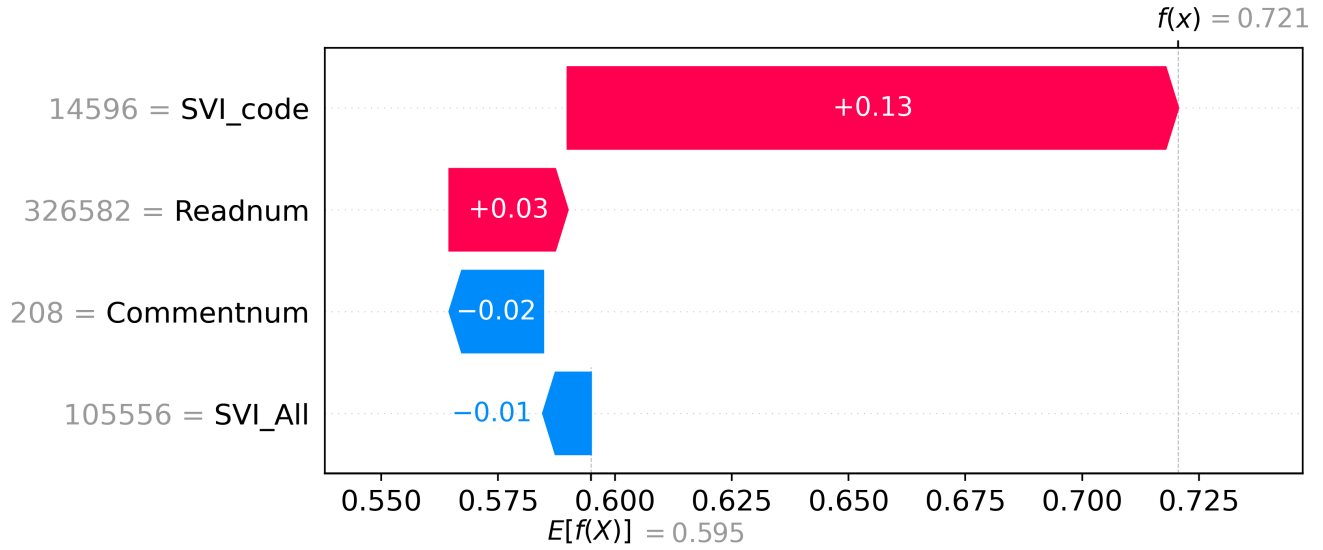


Figure 4: SHAP Values Waterfall Figure

Overall, the stock code search index and the number of reads on Guba have a significant positive impact on stock returns when yields are positive, reflecting the attention of professional and active investors in the market. The negative impact of the number of comments on Guba suggests that market controversy may lead to negative returns when yields are negative, which is consistent with the Behavioral Inhibitory System (BIS), which tends to avoid risk and negative consequences. In contrast, the positive effects of the stock code search index and the number of reads on Guba reflect the role of the Behavioral Incentive System (BAS), which encourages the pursuit of rewards and positive outcomes, thereby driving stock returns up when yields are positive (Muehlfeld, Weitzel and Witteloostuijn, 2013). Taken together, these results provide a better understanding of how positive and negative changes in stock returns affect market attention and its predictive role on stock returns through the psychological mechanisms of market participants (BIS and BAS).

5 Discussion

Study 1 verified hypothesis 1 that investment returns affect investors' behavioral inhibition, which in turn reduces attention to stock information. According to the Behavioral Inhibition System (BIS) theory, negative investment returns trigger investors' avoidance behavior and reduce their attention to stock information to reduce psychological stress. This finding is consistent with Gray (1982) theory, which suggests that during poor market performance, investors tend to avoid market information that may bring further negative emotions. This behavior is not only a reflection of the psychological defense mechanism but also illustrates the cognitive bias of investors in the face of risk.

Second, hypothesis 2 proposes that when investment returns are consistently negative or the market is generally depressed, investor attention fluctuates due to the alternating effects of the BIS and BAS systems. It was found that investors' market attention does fluctuate as market conditions deteriorate, which is consistent with the theory that the BAS system incentivizes investors to seek new opportunities in depressed markets (Muehlfeld, Weitzel and Witteloostuijn, 2013). This volatility reflects the psychological complexity of investors in the face of persistent negative returns, both the tendency to avoid risk and the

incentive to seek rebound opportunities in downturns. The practical implication is that understanding this behavioral pattern can help market regulators and financial service providers better predict market volatility and design interventions accordingly. Third, hypothesis 3 analyzes the predictive effect of return changes on investor behavior using machine learning models. It is found that the AdaBoost model performs best in predicting changes in investor behavior, and the SHAP value analysis reveals the impact of each feature on the model output. This result not only validates the value of applying machine learning methods in behavioral finance research but also demonstrates the differences in the behaviors of different types of market participants in the face of positive and negative returns. For example, the stock code search index and the number of stock bar readings have a significant positive effect on positive returns, indicating the high concern of professional investors and active market participants when the market returns are favorable, while the number of stock bar comments has a significant negative effect on negative returns, reflecting the negative emotional effect when the market controversy is high.

6 Conclusion

6.1 Limitations and Future Research

Due to time and word count constraints, this study has several limitations. In behavior-related information systems research, it is imperative to include at least one experimental component. This study, however, did not incorporate web-based experiments or real-person survey experiments that address actual human behavioral situations through questionnaire design and the utilization of psychological scales from the BIS and the BAS. Such experiments will be included in future research. Furthermore, the word limit of this assessment has led to the omission of substantial theoretical content in the Hypothesis Development and Methodology sections, and the Literature Review section has been condensed. The theoretical derivation of the hypotheses necessitates a more detailed explanation of how BIS and BAS theories specifically apply to the logical process of investor behavior. Additionally, the data limitations must be scrutinized, as there may be other potentially significant variables that have been overlooked.

6.2 Brief Conclusion

Based on the Behavioral Inhibition System (BIS) and Behavioral Activation System (BAS) theories, this study examines the impact of investment returns on retail investors' market attention behavior. By analyzing data from the CSI 300, the most representative index of China's stock market from 2014 to 2016, the study verifies three hypotheses and provides a corresponding discussion. Utilizing the theoretical framework of Behavioral Finance, along with statistical analysis and machine learning models, this research delves into how emotional returns influence investor behavior and market attention. The findings not only validate the hypotheses but also uncover the psychological mechanisms of market participants when faced with varying investment returns, highlighting their predictive role in market behavior.

References

- Agarwal, S., S. Kumar and U. Goel (2019). “Stock market response to information diffusion through internet sources: A literature review”. *International Journal of Information Management* 45, pp. 118–131.
- Arnold, M., M. Pelster and M. G. Subrahmanyam (2022). “Attention triggers and investors’ risk-taking”. *Journal of Financial Economics* 143.2, pp. 846–875.
- Aydınlıyurt, E. T., N. Taşkın, S. Scahill and A. Toker (2021). “Continuance intention in gamified mobile applications: A study of behavioral inhibition and activation systems”. *International Journal of Information Management* 61, p. 102414.
- Chai, D., M. Dai, P. Gharghori and B. Hong (2021). “Internet Search Intensity and Its Relation with Trading Activity and Stock Returns*”. *International Review of Finance* 21.1, pp. 282–311.
- Chen, T., Z. Gao, J. He, W. Jiang and W. Xiong (2019). “Daily price limits and destructive market behavior”. *Journal of Econometrics* 208.1, pp. 249–264.
- Chu, G., J. W. Goodell, X. Li and Y. Zhang (2023). “Understanding short-term price pressure from index reconstitutions: Evidence from the CSI 300”. *Accounting & Finance* 63.2, pp. 2421–2440.
- Daniel, K., D. Hirshleifer and S. H. Teoh (2002). “Investor psychology in capital markets: evidence and policy implications”. *Journal of Monetary Economics* 49.1, pp. 139–209.
- Deng, S., Z. Huang, A. P. Sinha and H. Zhao (2018). “The Interaction Between Microblog Sentiment and Stock Returns: An Empirical Examination”. *MIS Quarterly* 42.3, pp. 895–918, A1–A13.
- Drake, M. S., J. Jennings, D. T. Roulstone and J. R. Thornock (2017). “The Comovement of Investor Attention”. *Management Science* 63.9, pp. 2847–2867.
- Fama, E. F. and K. R. French (1988). “Dividend yields and expected stock returns”. *Journal of Financial Economics* 22.1, pp. 3–25.
- Fodor, J. A. and Z. W. Pylyshyn (1988). “Connectionism and cognitive architecture: A critical analysis”. *Cognition* 28.1, pp. 3–71.
- Frydman, C. and C. F. Camerer (2016). “The Psychology and Neuroscience of Financial Decision Making”. *Trends in Cognitive Sciences* 20.9, pp. 661–675.
- Gray, J. A. (1982). “Précis of The neuropsychology of anxiety: An enquiry into the functions of the septo-hippocampal system”. *Behavioral and Brain Sciences* 5.3, pp. 469–484.
- Grinblatt, M. and M. Keloharju (2000). “The investment behavior and performance of various investor types: a study of Finland’s unique data set”. *Journal of Financial Economics* 55.1, pp. 43–67.
- Hvide, H. K., T. G. Meling, M. Mogstad and O. L. Vestad (2024). “Broadband Internet and the Stock Market Investments of Individual Investors”. *The Journal of Finance* 79.3, pp. 2163–2194.
- Jin, X. and Q. Xue (2023). “Retail investor attention and stock market behavior in Russia-Ukraine conflict based on Chinese practices: Evidence from transfer entropy causal network”. *Finance Research Letters* 58, p. 104457.
- Lee, W. Y., C. X. Jiang and D. C. Indro (2002). “Stock market volatility, excess returns, and the role of investor sentiment”. *Journal of Banking Finance* 26.12, pp. 2277–2299.
- Li, Z., J. Liu, X. Liu and C. Wu (2024). “Investor attention and stock price efficiency: Evidence from quasi-natural experiments in China”. *Financial Management* 53.1, pp. 175–225.
- Lundberg, S. M. and S.-I. Lee (2017). “A Unified Approach to Interpreting Model Predictions”. *Advances in Neural Information Processing Systems*. Vol. 30.
- Muehlfeld, K., U. Weitzel and A. V. Witteloostuijn (2013). “Fight or freeze? Individual differences in investors’ motivational systems and trading in experimental asset markets”. *Journal of Economic Psychology* 34, pp. 195–209.
- Qiu, L. and I. Welch (2004). *Investor Sentiment Measures*. Working Paper 10794. National Bureau of Economic Research.
- Reyes, T. (2019). “Negativity Bias in Attention Allocation: Retail Investors’ Reaction to Stock Returns”. *International Review of Finance* 19.1, pp. 155–189.

- Richards, D. W., M. Fenton-O’Creevy, J. Rutterford and D. G. Kodwani (2018). “Is the disposition effect related to investors’ reliance on System 1 and System 2 processes or their strategy of emotion regulation?” *Journal of Economic Psychology* 66, pp. 79–92.
- Rijn, J. N. van and F. Hutter (2018). “Hyperparameter Importance Across Datasets”. *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. KDD ’18, pp. 2367–2376.
- Rodríguez-Pereira, J., B. Balcik, M.-È. Rancourt and G. Laporte (2021). “A Cost-Sharing Mechanism for Multi-Country Partnerships in Disaster Preparedness”. *Production and Operations Management* 30.12, pp. 4541–4565.
- Sun, Y., X. Liu, G. Chen, Y. Hao and Z. Zhang (2020). “How mood affects the stock market: Empirical evidence from microblogs”. *Information Management* 57.5, p. 103181.
- Tetlock, P. C. (2007). “Giving Content to Investor Sentiment: The Role of Media in the Stock Market”. *The Journal of Finance* 62.3, pp. 1139–1168.
- Wang, X., Q. Ye, F. Zhao and Y. Kou (2018). “Investor sentiment and the Chinese index futures market: Evidence from the internet search”. *Journal of Futures Markets* 38.4, pp. 468–477.
- Xian Li, J. A. H. and J. L. Teall (2016). “Investor Attention on the Social Web”. *Journal of Behavioral Finance* 17.1, pp. 45–59.
- Yang, L. and A. Shami (2020). “On hyperparameter optimization of machine learning algorithms: Theory and practice”. *Neurocomputing* 415, pp. 295–316.
- Zhang, X. and L. Zhang (2015). “How Does the Internet Affect the Financial Market? An Equilibrium Model of Internet-Facilitated Feedback Trading”. *MIS Quarterly* 39.1, pp. 17–38.
- Zhang, Y., G. Chu and D. Shen (2021). “The role of investor attention in predicting stock prices: The long short-term memory networks perspective”. *Finance Research Letters* 38, p. 101484.
- Zhao, S., X. Chen and J. Zhang (2019). “The systemic risk of China’s stock market during the crashes in 2008 and 2015”. *Physica A: Statistical Mechanics and its Applications* 520, pp. 161–177.
- Zhu, Z., L. Sun, K. Yung and M. Chen (2020). “Limited investor attention, relative fundamental strength, and the cross-section of stock returns”. *The British Accounting Review* 52.4, p. 100859.

Appendices

The code and data are available on GitHub

<https://github.com/bobbyZhong10/Research-on-Retail-Investor-sentiment-and-attention.git>

Appendix A

Metric	Readnum	Commentnum	SVI_code	SVI_All
Mean	797,961	991.24	8,438.19	42,740.47
Std	1,613,125	2,471.37	12,414.52	94,999.65
Min	0	0	0	769
Max	142,909,600	209,996	787,205	3,103,777

Table 7: Statistical Metrics for DV

Industry	Count	Industry	Count
Special Equipment Manufacturing	12	Coal Mining and Washing	16
Internet and Related Services	8	Gas Production and Supply	2
Accommodation	1	Ecological Protection and Environmental Governance	2
Insurance	5	Telecommunications, Broadcasting, TV, and Satellite Transmission	7
Public Facilities Management	1	Electricity, Heat Production, and Supply	17
Other Financial Industries	2	Electrical Machinery and Equipment Manufacturing	11
Agriculture	3	Animal Husbandry	2
Agricultural and Sideline Food Processing	4	Petroleum Processing, Coking, and Nuclear Fuel Processing	2
Chemical Raw Materials and Chemical Products	10	Oil and Gas Extraction	3
Chemical Fiber Manufacturing	1	Leasing	1
Pharmaceutical Manufacturing	28	Textile and Apparel Manufacturing	3
Health	1	General	3
Business Services	8	Air Transport	6
Civil Engineering Construction	16	Computer, Communication, and Other Electronic Equipment Manufacturing	29
Broadcasting, TV, Film, and Recording Production	5	Currency and Financial Services	15
Building Decoration and Other Construction	2	Capital Market Services	26
Mining Auxiliary Activities	3	Software and Information Technology Services	18
Real Estate	20	General Equipment Manufacturing	5
Wholesale	9	Road Transport	1
Cultural and Artistic Industries	1	Alcohol, Beverage, and Refined Tea Manufacturing	10
Cultural, Educational, and Sports Goods Manufacturing	1	Metal Products	2
News and Publishing	6	Railway, Ship, Aerospace, and Other Transport Equipment Manufacturing	12
Non-ferrous Metal Smelting and Rolling	10	Railway Transport	2
Non-ferrous Metal Mining	9	Retail	10
Rubber and Plastic Products	2	Non-metallic Mineral Products	6
Water Transportation	7	Food Manufacturing	5
Water Production and Supply	3	Ferrous Metal Smelting and Rolling	6
Automobile Manufacturing	14	Ferrous Metal Mining	1

Table 8: Industry Counts

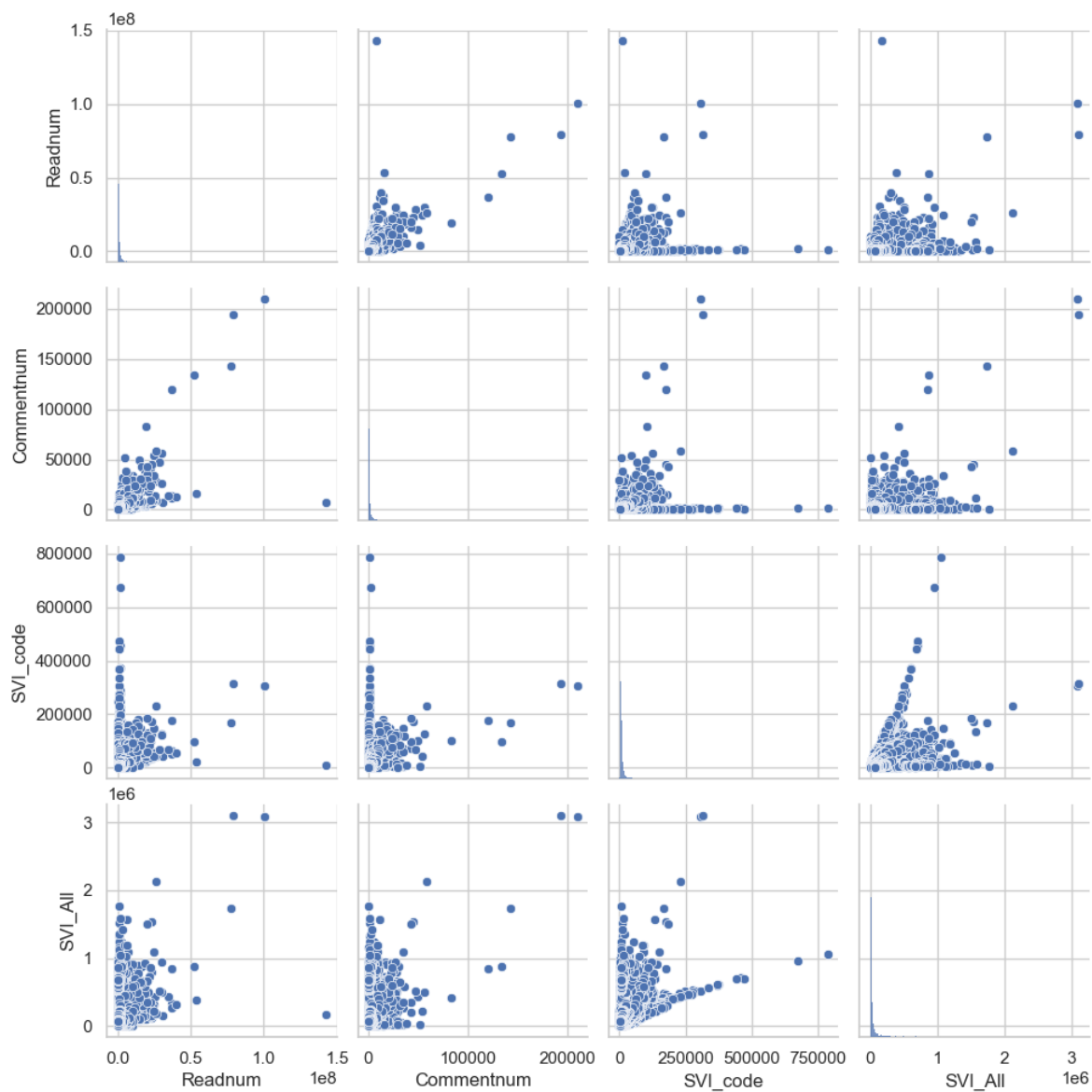


Figure 5: Relationship between variables

Appendix B

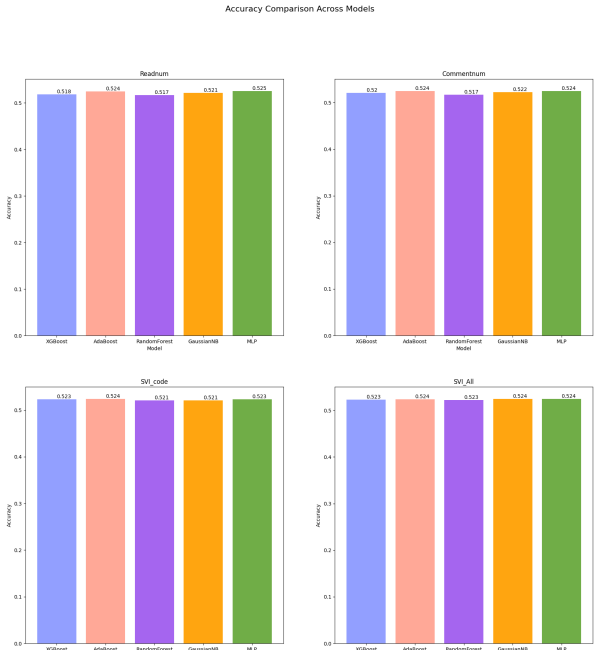


Figure 6: Study 3 Accuracy

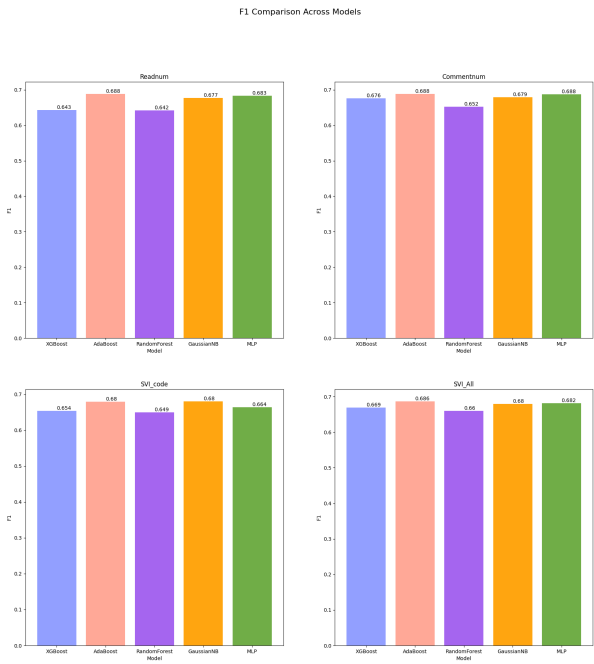


Figure 7: Study 3 F1 Score

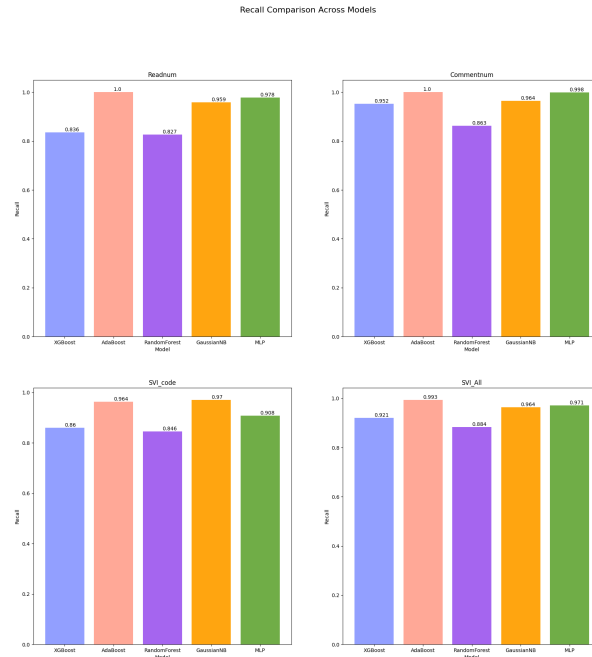


Figure 8: Study 3 Recall